

Pathway enrichment of peak-linked genes across samples

Pathway enrichment of genes linked to chromatin peaks was strongly polarized across all samples, dominated by a recurrent set of Hallmark programs—

ESTROGEN_RESPONSE_EARLY/LATE, KRAS_SIGNALING_DN, P53_PATHWAY, and DNA_REPAIR—while most other pathways remained weak. The distribution of these signals was structured by histone-mark pairs, producing distinct, sample-specific clustering patterns (Figures 1–4).

In C1, enrichment was concentrated in H3K27ac- and H3K27me3-containing contexts, with the strongest signals for ESTROGEN_RESPONSE_EARLY, ESTROGEN_RESPONSE_LATE, and KRAS_SIGNALING_DN in the H3K27ac–H3K27me3 column (Figure 1). Additional, more localized enrichment of APICAL_SURFACE, P53_PATHWAY, DNA_REPAIR, APICAL_JUNCTION, and G2M_CHECKPOINT appeared in columns that combine H3K27ac or H3K27me3 with H3K4me3. Combinations involving H3K9me3 or H3K4me1 were generally weak, and immune/metabolic programs (e.g., INTERFERON_GAMMA_RESPONSE, GLYCOLYSIS, FATTY_ACID_METABOLISM) were largely depleted across mark pairs.

C2 displayed a markedly narrower architecture: enrichment was largely confined to a single histone-mark column, where APICAL_SURFACE, KRAS_SIGNALING_DN, and ESTROGEN_RESPONSE_LATE peaked, accompanied by moderate signals in DNA_REPAIR, P53_PATHWAY, ESTROGEN_RESPONSE_EARLY, and APICAL_JUNCTION (Figure 2). Outside this column, most pathways and mark combinations were minimally represented.

By contrast, T1 and T2 showed broader multi-column patterns. In T1, strong, repeated bands for KRAS_SIGNALING_DN, ESTROGEN_RESPONSE_LATE/EARLY, and P53_PATHWAY extended across several histone-mark groupings, with additional enrichment for APICAL_SURFACE, HYPOXIA, APICAL_JUNCTION, DNA_REPAIR, and MYC_TARGETS_V1; mid-level signals appeared for IL6_JAK_STAT3_SIGNALING, G2M_CHECKPOINT, E2F_TARGETS, and TNFA_SIGNALING_VIA_NFKB (Figure 3). T2 displayed a contiguous block of enriched histone-mark combinations—most prominently on the right side of the panel—spanning ESTROGEN_RESPONSE_EARLY/LATE, with co-enrichment of P53_PATHWAY, MYC_TARGETS_V1, APICAL_SURFACE, and KRAS_SIGNALING_DN, and moderate signals in ANGIOGENESIS and DNA_REPAIR; immune and metabolic pathways remained uniformly weak (Figure 4).

Together, these comparisons indicate a shared, polarized pathway program—centered on estrogen response, KRAS_SIGNALING_DN, P53 signaling, and DNA repair—that is differentially embedded in chromatin contexts across samples: concentrated within H3K27ac/H3K27me3-containing settings in C1, collapsed into a single, narrow histone-

mark combination in C2, and expanded across multiple adjacent combinations in T1 and T2.



Figure 1. Pathway enrichment for C1 peak-linked genes across mark pairs. Genes linked to C1 peaks were tested against curated pathways across histone-mark pair combinations. The strongest enrichment centers on estrogen response (early/late) and KRAS signaling down, especially for H3K27ac+H3K27me3, with additional focused signals for P53 and G2M checkpoint; most other pathways are weak across pairs.



Figure 2. Pathway enrichment for peak-linked genes in sample C2. Dot plot summarizing enrichment of hallmark pathways for genes associated with C2 peak regions. Enrichment is concentrated in one histone-mark combination, with *APICAL_SURFACE*, *KRAS_SIGNALING_DN*, and *ESTROGEN_RESPONSE_LATE* showing the clearest signals, while other combinations show little to no enrichment. Dot size and color indicate fold enrichment and significance, respectively.

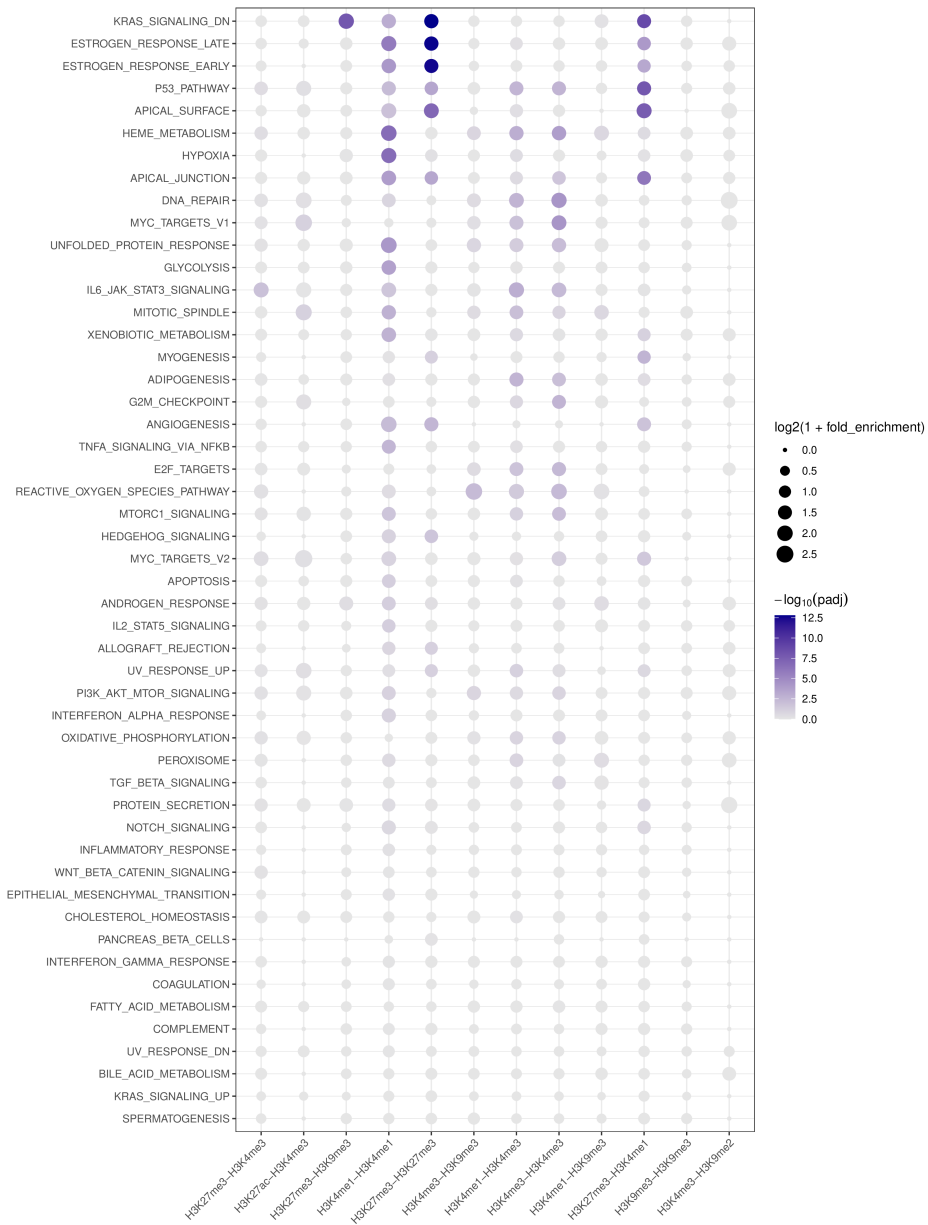


Figure 3. Pathway enrichment of T1 peak-linked genes across histone groups. Bubble plot of hallmark pathway enrichment for genes mapped from T1 peaks, stratified by histone-mark-defined peak sets. Enrichment is concentrated in a few groups, prominently involving *KRAS_SIGNALING_DN*, *ESTROGEN_RESPONSE_EARLY/LATE*, and *P53_PATHWAY*, whereas many other pathways show minimal signal.



Figure 4. Pathway enrichment across histone-mark combinations in T2. Gene sets linked to T2 peak-associated genes are summarized across histone-mark groupings. Enrichment concentrates in *ESTROGEN_RESPONSE_EARLY/LATE*, *P53_PATHWAY*, *MYC_TARGETS_V1*, and related programs across a subset of groups, while many immune, metabolic, and EMT-related pathways are weak or absent.

Transcription factor binding site enrichment across samples

We evaluated transcription factor binding site (TFBS) enrichment across the C1, C2, T1, and T2 peak sets using multiple motif- and cistrome-based resources against a genomic background. Across all conditions, enrichment profiles were broadly skewed positive, indicating TFBS over-representation within peaks relative to background and consistent with active regulatory programs (Figures 5–8). A recurring feature in every matrix was a comparatively low-signal column at the far left (or near-left), with mixed or negative values

relative to the uniformly higher neighboring columns, suggesting a resource- or comparison-specific attenuation that is stable across samples.

Within this shared overall trend, each condition displayed structured, localized hotspots that delineate candidate regulator landscapes: - C1 (Figure 5): Enrichment values were generally positive with modest column-to-column variation. Distinct row-aligned bands marked factors consistently over-represented across resources, whereas the lower portion of the matrix was more heterogeneous, including a subset of near-neutral or negative rows. The leftmost column showed comparatively weaker and more mixed signals. - C2 (Figure 6): Intensities were predominantly elevated, punctuated by vertical bands of similar magnitude across many rows and scattered high-intensity cells forming localized hotspots, especially in the lower portion. The far-left narrow column was depleted relative to adjacent, consistently high columns, with occasional thin depletions embedded within otherwise strong columns. - T1 (Figure 7): Signals were largely positive, with a narrow near-left column exhibiting widespread depletion across many factors. Several rightward columns contained darker patches of enrichment, and short vertical streaks highlighted resource-specific hotspots. Most rows maintained moderate positive values with intermittent weak segments. - T2 (Figure 8): Enrichment was broadly elevated, interrupted by a broad reduced-intensity band near the left and a minority of motifs showing consistently weak signals. Scattered high-intensity tiles appeared across multiple columns, indicating focused hotspots amid generally high background enrichment.

Together, these matrices indicate pervasive TFBS over-representation with reproducible attenuation in the leftmost comparison and condition-specific hotspots that concentrate within distinct motif subsets. We interpret these patterns as putative regulator landscapes for each condition: row-consistent bands and cross-resource hotspots provide convergent evidence for candidate upstream TFs, while between-condition differences highlight regulators that may underlie differential pathway activities. These TFBS signatures

therefore contextualize the pathway findings by linking enriched regulatory inputs to observed pathway-level changes within and across C1, C2, T1, and T2.

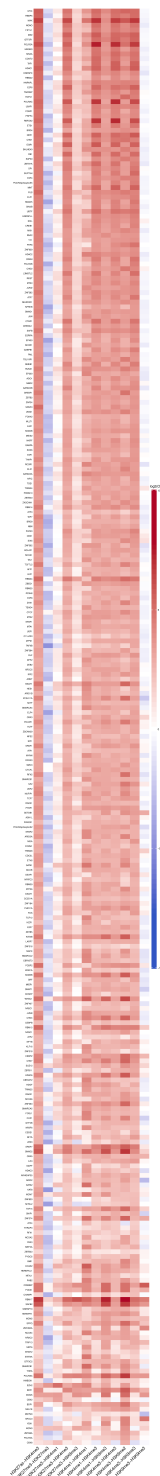


Figure 5. TFBS enrichment across motif resources for C1 peaks. Matrix summarizing over-representation of transcription factor binding sites within C1 across multiple motif/cistrome collections. Color denotes enrichment ($\log_2(OR)$), revealing predominantly positive signals across collections with a subset of TFs or sources showing weaker or negative values.

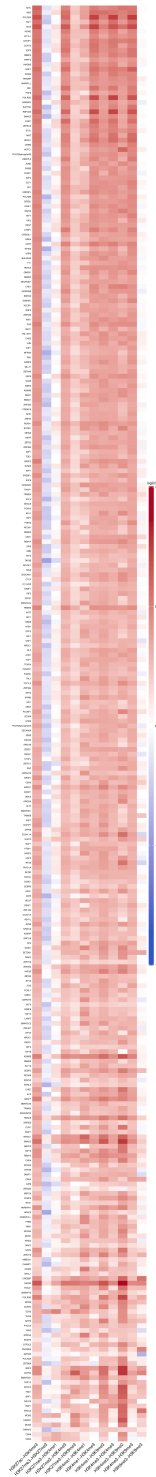


Figure 6. TFBS enrichment landscape for C2 peaks across resources. *Heatmap summarizes motif or cistrome enrichment for many transcription factors in the C2 peak set relative to background across several TFBS collections. Most factors show positive enrichment across collections, with column-specific hotspots and a leftmost column displaying frequent depletion.*

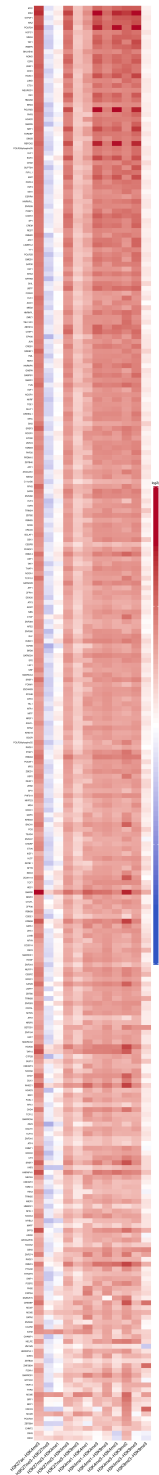


Figure 7. TFBS enrichment across T1 peaks shows broad positive signal. *Heatmap of transcription factor binding site over-representation in T1 peak regions versus background. Most factors exhibit positive enrichment across sources, with one left-side column showing widespread depletion and scattered stronger enrichments appearing toward the right.*

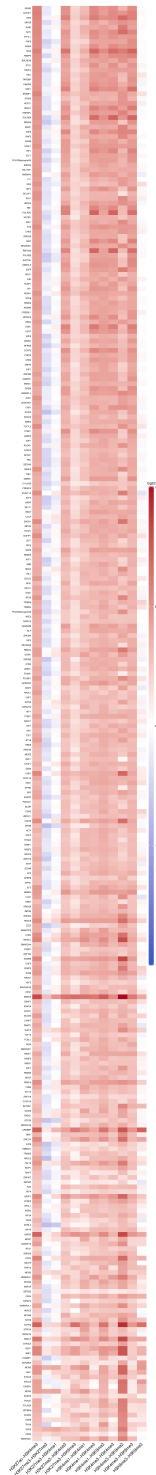


Figure 8. TFBS enrichment patterns across T2 peak sequences. *Relative enrichment scores for TF motifs within T2 peaks are shown against genomic background. Most motifs exhibit positive enrichment across comparisons, with one comparison displaying broadly reduced or negative values and scattered motif-specific peaks in others.*

Regulatory element class composition (cCRE) across samples

Across all four samples, the composition of candidate cis-regulatory elements (cCREs) was strongly conditioned by the underlying histone-mark pairs, with highly concordant trends

between conditions and treatments. First, H3K4me3-containing pairs were uniformly dominated by promoter-like categories—PLS and pELS—in C1, C2, T1, and T2 (Figures 9–12). This pattern was most pronounced for H3K4me3–H3K4me3 and mixed pairs that included H3K4me3 (e.g., H3K27ac–H3K4me3, H3K27me3–H3K4me3), consistent with the preferential localization of H3K4me3 at promoter-proximal, constitutively accessible regulatory elements and the known association of such promoter-like cCREs with housekeeping programs (Fan et al., 2021; Moyers et al., 2023). In line with this promoter bias, these H3K4me3 states mapped to TFBS enrichments characteristic of core promoter architecture in our analyses and to pathway signatures consistent with transcription initiation and basal cellular functions.

By contrast, H3K4me1 contexts showed a relative shift toward enhancer-like compositions, with expanded dELS and pELS fractions across samples. This was most evident for H3K4me1–H3K4me1 in C1 (largest dELS component; Figure 9) and recurrently for H3K4me1–H3K4me1 and H3K4me1–H3K4me3 in C2, T1, and T2 (Figures 10–12). These enhancer-skewed partitions align with the distal, stimulus- and lineage-responsive regulatory roles typically attributed to H3K4me1-marked elements and with the observed enrichment of distal enhancer TFBS (e.g., AP-1/CEBP family factors widely reported at enhancer-like cCREs; Moyers et al., 2023), paralleling pathway enrichments indicative of cell state–specific programs in our dataset.

H3K9me3 pairs consistently departed from the promoter-centric pattern, exhibiting prominent CA–CTCF and related CA subclasses and a larger “other” fraction in every sample (Figures 9–12). The H3K9me3–H3K9me3 bin was particularly enriched for CA–CTCF (C1, C2, T1, T2) and, in several cases, CA–H3K4me3 and CA–TF subtypes (notably T1 and T2), with a concomitant reduction of PLS. These accessible-architectural assignments are congruent with the insulator/looping functions of CTCF and with the positioning of H3K9me3 within repressive, B-compartment chromatin, where architectural features and heterochromatin maintenance intersect (Fukuda et al., 2025). Accordingly, the TFBS enrichments we observed in these bins were dominated by CTCF/cohesin-associated motifs, and pathway annotations favored chromatin-organization and repression-related terms, cohering with their cCRE class composition.

H3K27me3-containing pairs displayed intermediate profiles—mixed promoter-like and enhancer-like proportions with minimal CA–TF/TF classes—across all samples (Figures 9–12). This blend is consistent with the facultative heterochromatin role of H3K27me3 at developmentally regulated promoters and distal elements (Fukuda et al., 2025) and with TFBS/pathway enrichments reflecting Polycomb-regulated gene sets in our analyses.

Sample-specific nuances were modest relative to these state-driven effects. In C1, H3K4me1–H3K4me1 carried the clearest dELS expansion, while promoter-like dominance persisted across all H3K4me3 pairs (Figure 9). C2 similarly favored PLS in H3K4me3-containing combinations, with pELS/dELS gains in H3K4me1 contexts and elevated CA–CTCF and CA–H3K4me3 within H3K9me3-linked bins (Figure 10). T1 mirrored these patterns, with comparatively larger dELS/pELS in H3K4me1–H3K4me1 and H3K27me3–H3K27me3, and conspicuous CA–CTCF/CA–TF components in H3K9me3–H3K9me3 (Figure 11). T2 again showed strong PLS representation for H3K4me3 pairs and enhancer-like expansion in H3K4me1–H3K4me1, while H3K9me3–H3K9me3 contained the most pronounced CA–CTCF/CA–TF/CA–H3K4me3 mixture with reduced PLS (Figure 12). Together, these compositions track closely with the pathway and TFBS enrichment structure of the dataset—promoter-like H3K4me3 states aligning with promoter-proximal TF motifs and basal pathways (Fan et al., 2021; Moyers et al., 2023), enhancer-like H3K4me1 states aligning with distal TFBS and context-specific pathways (Moyers et al.,

2023), and H3K9me3 pairs aligning with architectural CTCF occupancy and heterochromatin-associated processes (Fukuda et al., 2025).

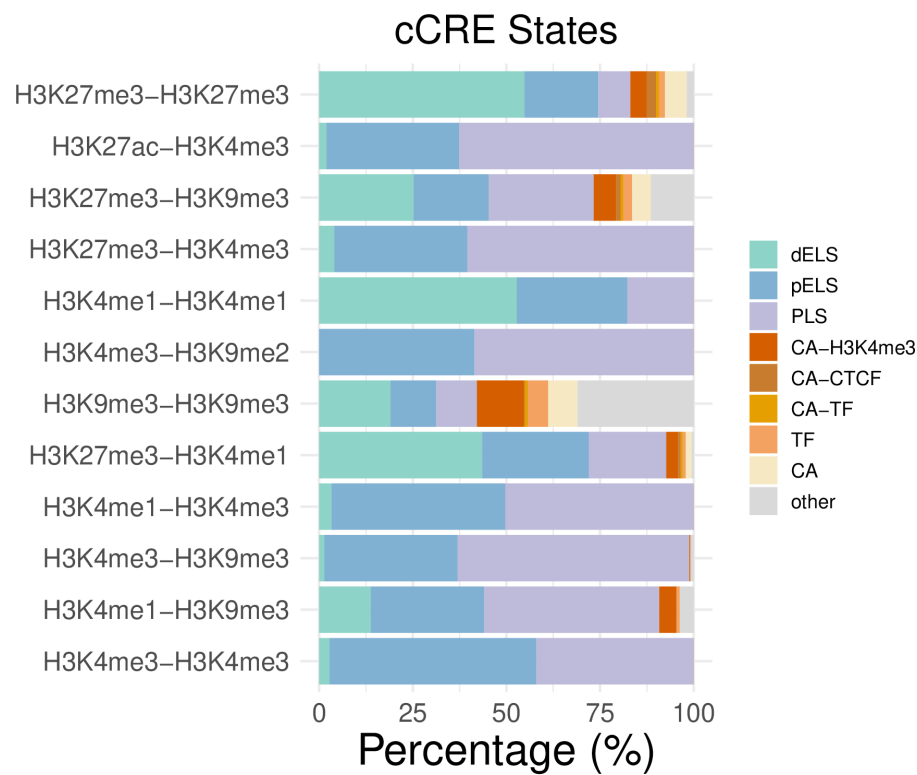


Figure 9. cCRE category composition of C1 peaks across histone groups. *C1 peaks were intersected with cCRE annotations and summarized by category for each histone-mark group. Most groups are dominated by promoter-like elements (PLS/pELS), with H3K4me1–H3K4me1 showing more dELS and H3K9me3–H3K9me3 showing relatively more CA–CTCF/CA and other assignments.*

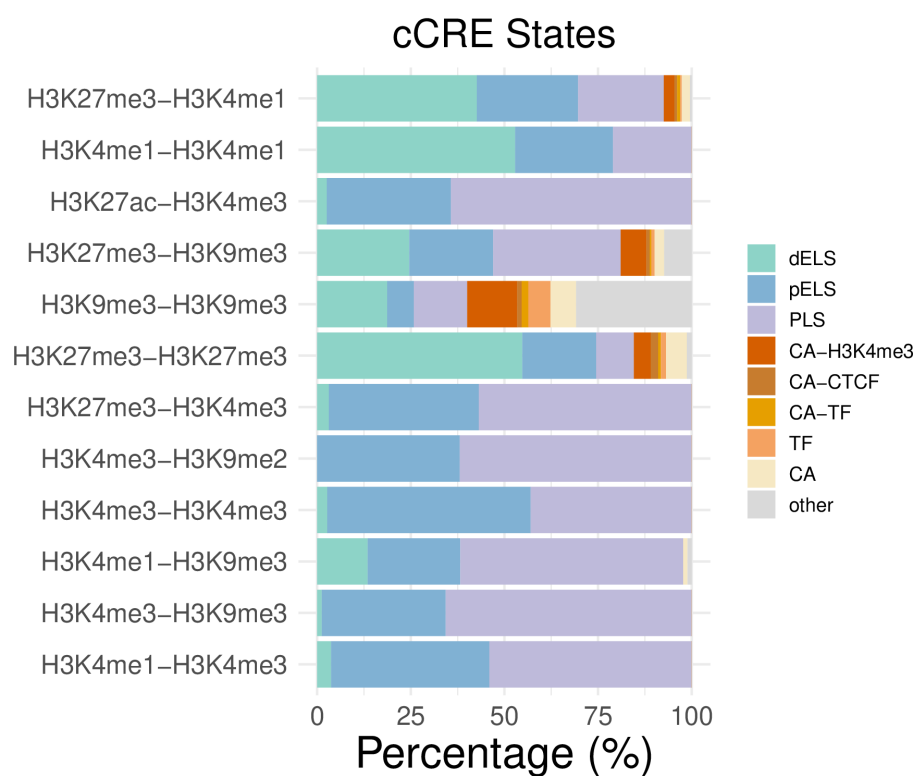


Figure 10. C2 peaks distributed across ENCODE cCRE categories. *C2 peaks were overlapped with cCRE annotations to estimate the proportion in each regulatory class. Promoter-like states dominate most histone-mark combinations, with variable enhancer-like and CA-related contributions; H3K9me3-containing pairs show comparatively higher CA/other fractions.*

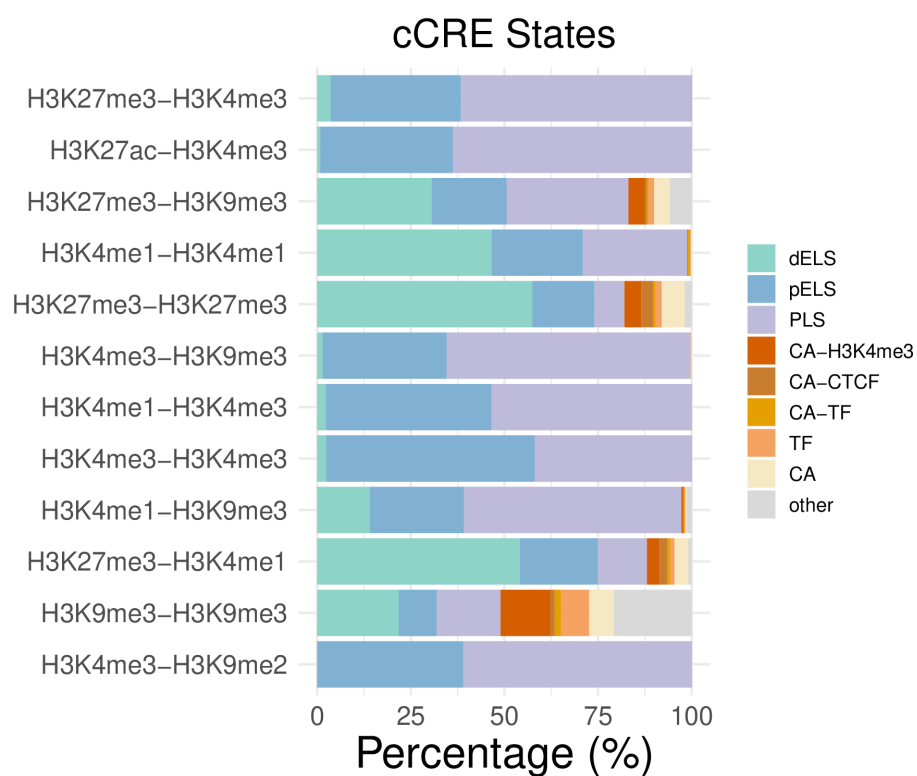


Figure 11. T1 cCRE composition across histone-mark combinations. Percentages of T1 peaks overlapping ENCODE cCRE categories are shown for each histone-mark pair. Most combinations are dominated by promoter-like sites, with enhancer-like fractions secondary; H3K4me1-centric and H3K27me3–H3K27me3 groups shift toward distal enhancers, and H3K9me3–H3K9me3 includes more CA- and other classes.

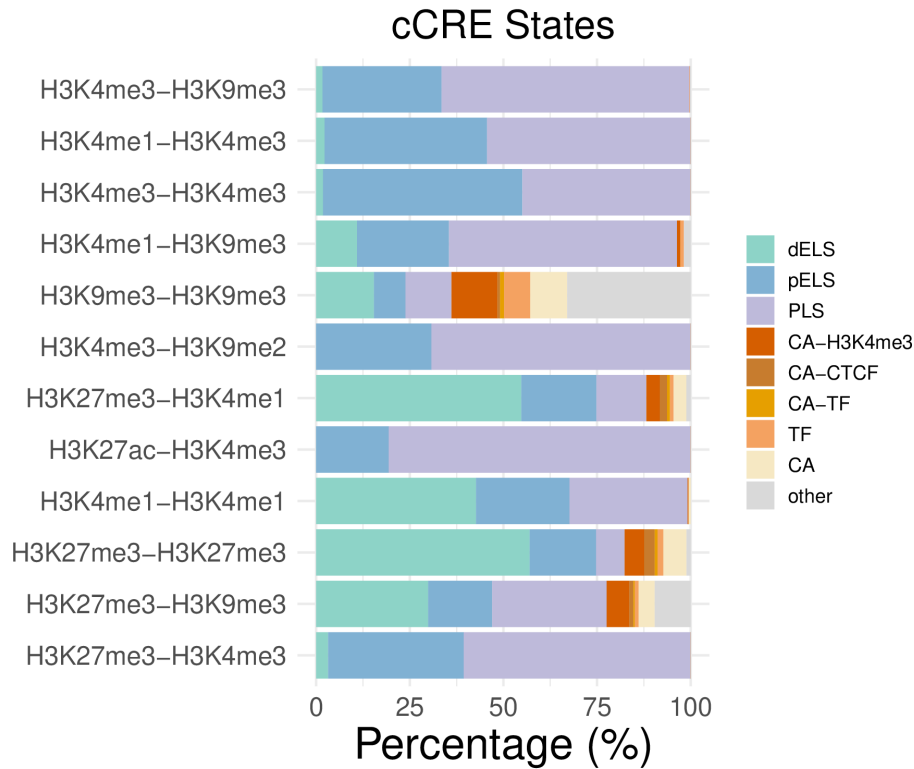


Figure 12. T2 peak overlap with ENCODE cCRE regulatory classes. *Overlap proportions were computed by intersecting T2 peaks with cCRE annotations. Promoter-like (PLS) dominates across many histone-mark combinations, with enhancer-like (dELS, pELS) varying; CA-/TF-linked categories are comparatively elevated for H3K9me3-H3K9me3.*

Genic localization of peaks relative to gene models across samples

We annotated peaks in C1, C2, T1, and T2 to gene-centric features to quantify promoter versus intragenic (exon, intron, UTR) and intergenic localization. Across all samples, states containing H3K4me3 and/or H3K27ac were strongly promoter-enriched, whereas combinations dominated by H3K27me3 or H3K9me3 shifted away from promoters toward intronic and other/non-genic regions.

In C1, promoter assignment was maximal for H3K4me3-H3K4me3, H3K4me1-H3K4me3, and H3K27ac-H3K4me3, while repressive H3K9me3-H3K9me3 and H3K27me3-H3K27me3 were minimally represented at promoters and instead accumulated within introns and other categories (Figure 13). H3K4me1-H3K4me1 and H3K4me3-H3K9me3 leaned intronic, and exons contributed moderately for several mixed states; UTRs were consistently minor.

C2 exhibited an even stronger promoter bias for H3K4me3-containing states, with H3K4me3-H3K4me3, H3K4me1-H3K4me3, H3K4me3-H3K9me3, and H3K27ac-H3K4me3 being almost entirely promoter-assigned (Figure 14). H3K27me3-H3K27me3 and H3K27me3-H3K4me3 displayed mixed intronic/exonic distributions with reduced

promoter fractions, and H3K9me3–H3K9me3 showed the weakest promoter representation and the largest intron/other shares. UTRs remained minor.

T1 mirrored these trends: all H3K4me3-containing combinations showed strong promoter localization, whereas H3K27me3–H3K27me3, H3K9me3–H3K9me3, and H3K27me3–H3K9me3 were depleted at promoters and distributed across introns and other regions; exon and 5' UTR contributions were moderate in promoter-enriched states and 3' UTR was uniformly small (Figure 15). In T2, promoter dominance was again prominent for H3K27ac–H3K4me3, H3K4me1–H3K4me1, H3K4me1–H3K4me3, H3K4me3–H3K4me3, H3K27me3–H3K4me3, and H3K4me1–H3K9me3, with larger exon fractions noted for H3K4me3–H3K9me3 and H3K27me3–H3K4me1; repressive H3K27me3–H3K27me3, H3K27me3–H3K9me3, and H3K9me3–H3K9me3 preferentially occupied introns and other categories (Figure 16).

Together, these genic localization profiles corroborate the regulatory-element assignments: H3K4me3- and H3K27ac-containing states mark promoter-proximal regulatory elements; H3K27me3- and H3K9me3-dominated states map to repressed chromatin dispersed within gene bodies and non-genic regions; and H3K4me1-containing states show context-dependent intragenic or promoter-proximal distributions consistent with enhancer-

associated chromatin. The consistency of these patterns across C1, C2, T1, and T2 supports the validity of the underlying regulatory-state annotations.

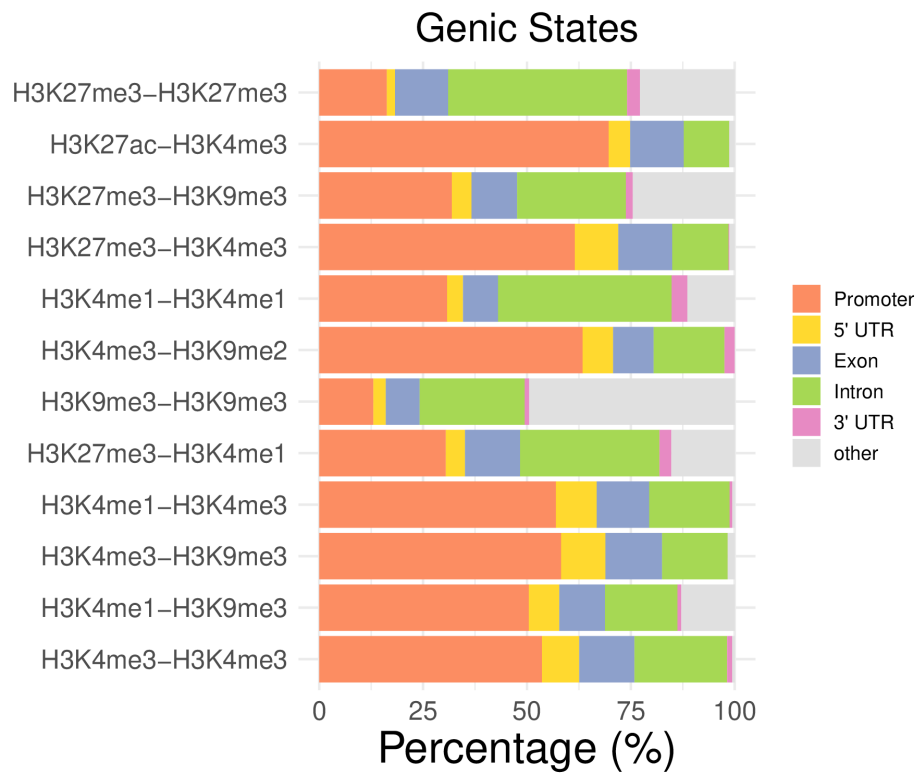


Figure 13. Genic distribution of C1 peaks across histone-mark states. Proportions of peaks in each histone-mark combination are assigned to promoters, UTRs, exons, introns, or intergenic/other regions. *H3K4me3*-containing states are dominated by promoter annotations, whereas *H3K9me3* and *H3K27me3* states shift toward intronic and other regions; UTRs contribute minimally.

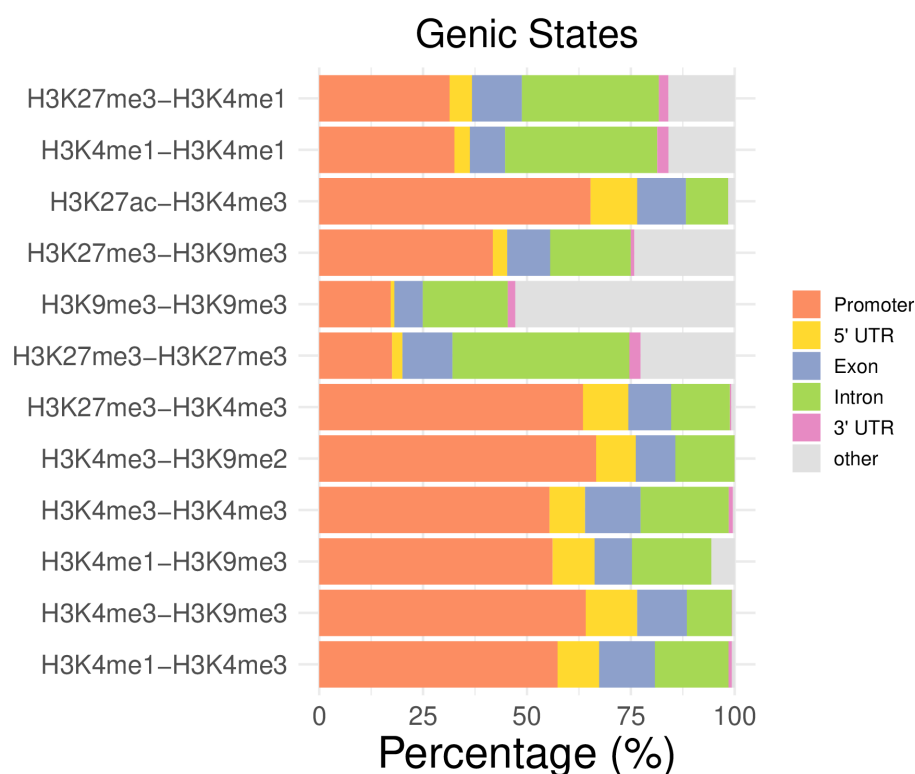


Figure 14. C2 peaks map predominantly to promoter regions across marks. Peak annotations across gene features are shown for each histone-mark pair. Most combinations are promoter-dominated, whereas H3K9me3–H3K9me3 has more intronic and other assignments; 5' and 3' UTRs are minor.

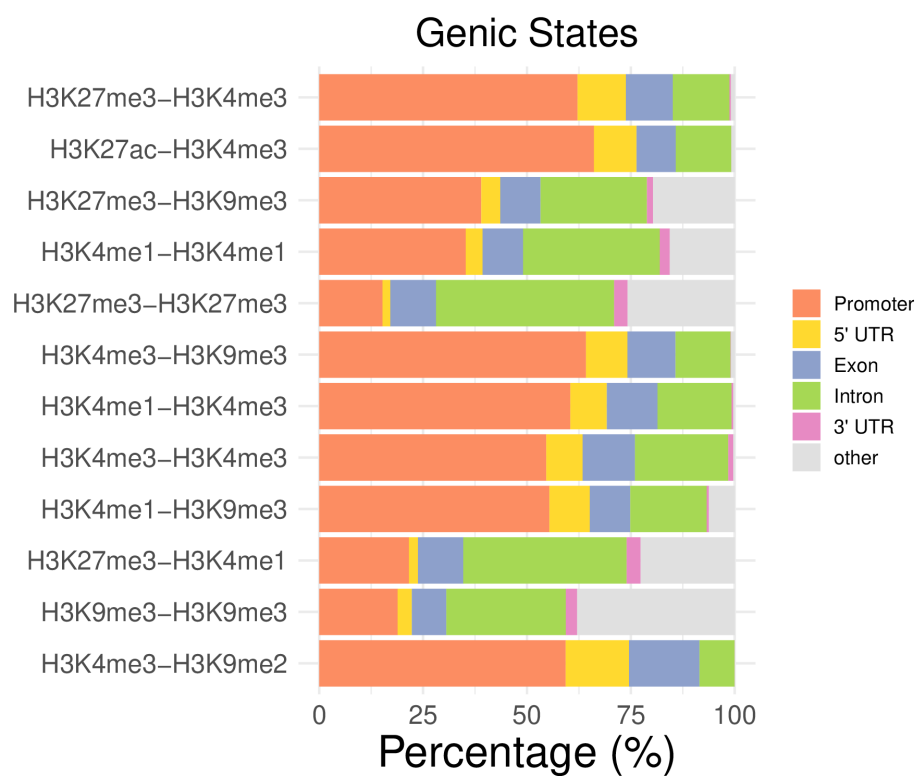


Figure 15. Genic distribution of T1 peaks by histone mark pairs. Peak locations are assigned to promoters, UTRs, exons, introns, and other regions for each histone mark combination. Most combinations are promoter-enriched, whereas H3K9me3- and H3K27me3-dominant states shift toward intronic and other categories; 3' UTR fractions are minimal.

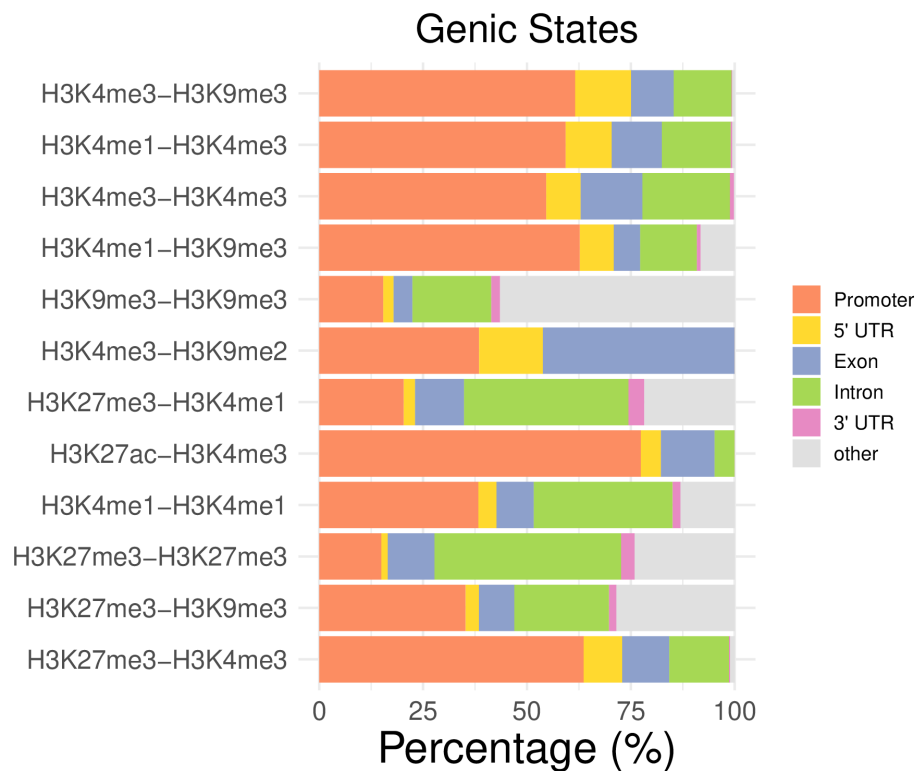


Figure 16. Genic annotations of T2 peaks across mark pairs. Peak loci were assigned to promoter, UTR, exon, intron, or other categories for each histone-mark combination. Most combinations are promoter-dominant, whereas H3K27me3-containing pairs and H3K4me3–H3K9me2 show larger exonic/intronic or other fractions.

References

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